

AURA V5 — Advanced Data Intelligence Report

This document presents a comprehensive analysis of the AURA V5 data system, including architecture, data fusion process, feature categories, and contribution of each intelligence layer.

1. System Overview

AURA V5 is a production-grade real estate intelligence infrastructure designed as a multi-layer spatio-temporal dataset. It integrates spatial, temporal, behavioral, macroeconomic, and market signals into a unified panel structure.

2. Dataset Characteristics

- Rows: 1,252,804
- Columns: 174
- Time Range: 2019 – 2025
- Granularity: grid_id_250m × municipality × year × quarter

The dataset is structured as a panel, enabling temporal learning and predictive modeling.

3. Fusion Pipeline Architecture

- STEP 1: Base Layer → Cat5 + Cat8 (grid-level join)
- STEP 2: Alignment → grid mapped to municipality
- STEP 3: Target Injection → Cat7 (municipality + time)
- STEP 4: Context Layers → Cat2, Cat3, Cat4, Cat1_2
- STEP 5: Behavioral Layer → Cat6
- STEP 6: Final Layer → Cat1 global profile

All joins are controlled to avoid explosion and ensure strict key alignment.

4. Spatial Intelligence (Cat5)

Represents urban morphology: - density - building footprint - infrastructure - sustainability metrics
Contribution: Defines physical structure and spatial constraints of properties.

5. Desirability Layer (Cat8)

Captures location attractiveness: - POI density - accessibility - proximity metrics
Contribution: Measures livability and attractiveness of each grid.

6. Target Layer (Cat7)

Target variable: price_per_m2
Characteristics: - Fully complete (0% missing) - Municipality-time based
Contribution: Ground truth for model training.

7. Market Dynamics (Cat2)

Includes: - price indices - market trends - liquidity indicators
Contribution: Captures how the market evolves over time.

8. Macro Layer (Cat3)

Includes macroeconomic signals: - inflation - economic regimes - global indicators Contribution: Explains external drivers of real estate markets.

9. Socioeconomic Layer (Cat4)

Includes: - population - housing supply - demographic structure Contribution: Represents demand-side fundamentals.

10. Micro Market Signals (Cat1_2)

Includes: - momentum - volatility - trend strength Contribution: Captures short-term market behavior.

11. Behavioral Signals (Cat6)

Includes: - tourism pressure - investor behavior - demand intensity Contribution: Reflects real-world usage and pressure on the market.

12. Global Property Layer (Cat1)

Cat1 was aggregated globally due to lack of spatial alignment. It provides average structural property characteristics. Contribution: Minimal predictive impact but maintains architectural completeness.

13. Data Quality Assessment

- Missing Values: 0%
- Duplicates: 0
- Infinite Values: 0

The dataset meets production-grade data quality standards.

14. Strengths and Limitations

Strengths: - Multi-layer intelligence - Clean data - Scalable design Limitations: - Partial municipality alignment - No transaction-level granularity

15. Final Conclusion

AURA V5 represents a complete real estate intelligence system ready for machine learning deployment. It integrates multiple dimensions of information into a single coherent framework.